

6. Time-Varying Volatility

In contrast to the weak predictability of equity risk premiums, there is a great deal of predictability in aggregate market volatilities.³¹

First, the volatility of equities is relatively high and much higher than *fundamental* volatilities of dividends (and also of consumption as a factor, as mentioned in section 2). This is labeled “*excess volatility*” by Robert Shiller (1981). From January 1935 to December 2010, the standard deviation of log returns is 16.0%. In contrast, the standard deviation of dividend growth is 9.6%, and thus return volatility is higher than fundamental volatility. However, taking earnings as fundamentals gives a different story. The volatility of earnings growth is 36.0%, and there is no excess volatility puzzle. But earnings cannot be directly used to value equity because they are generated by a firm owned by both debt and equity holders (it is also an accounting construct and not an actual cash flow payment).³² Dividends accrue only to stockholders. The fact that equity return volatility is higher than dividend volatility is a reflection that some predictability of risk premiums is coming from discount rates and not from cash flows in the Cochrane versus Bansal–Yaron debate.

Models to predict volatility were created by Robert Engle (1982) and are called ARCH models (for “autoregressive conditional heteroskedasticity”). This is a mouthful and means that conditional volatility changes over time (“heteroskedasticity” is Greek for “different dispersions”) in such a way that it is mean-reverting (“autoregressive”). The model was extended by Bollerslev (1986) to GARCH, for “generalized autoregressive conditional heteroskedasticity,” and that is the name used by industry and the literature. The GARCH model revolutionized volatility modeling, especially for risk management. Engle was awarded the Nobel Prize in 2003.

In the basic GARCH model, conditional variance, σ_t^2 , follows:³³

$$\sigma_t^2 = a + b\sigma_{t-1}^2 + c\varepsilon_{t-1}^2. \quad (8.8)$$

The main effect in equation (8.8) is that volatility depends on itself and is thus persistent (through b , which is close to one), or autoregressive, and it is also affected by past shocks (through c). Thus, large shocks—like those in 1987 (stock market crash), 1998 (Russian default and emerging markets crisis), and 2008

³¹ Most of this predictability in equity volatility comes from equity volatility predicting itself. There is little predictive ability of macro variables to forecast market volatility, as Paye (2012) shows.

³² There are many earnings models, but one seminal paper that links the dividend discount model with what are now called *residual income models* is Miller and Modigliani (1961). These are the same Miller and Modigliani who received the Nobel prize in 1985 for (the irrelevance of) capital structure. My paper on this is Ang and Liu (2001).

³³ There have been many extensions in the literature for different past volatility dependence, asymmetry, including jumps, and so on.